

TOI-1811b

R-2219

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1811_250318 · created 2026-08-14T23:37:43+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460752.676 +/- 0.018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.048
Transit depth (Rp/Rs)^2	2.67 +/- 1.55 [%]
Orbital Inclination (inc)	89.26 +/- 2.47
Airmass coefficient 1 (a1)	0.52 +/- 0.26
Airmass coefficient 2 (a2)	0.15 +/- 0.26
Scatter in the residuals of the lightcurve fit is	45.55 %
Best Comparison Star	None
Optimal Aperture	6.98
Optimal Annulus	8.58
Transit Duration (day)	0.087 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.163 ± 0.048 vs 0.13272 (deviation 0.63σ)
Duration fit vs published	0.087 d vs 0.0813 d (deviation 0.27σ)
Mid-time O–C	-22.6 min
Depth SNR	3.4
Residual scatter	45.55 %

Light curve

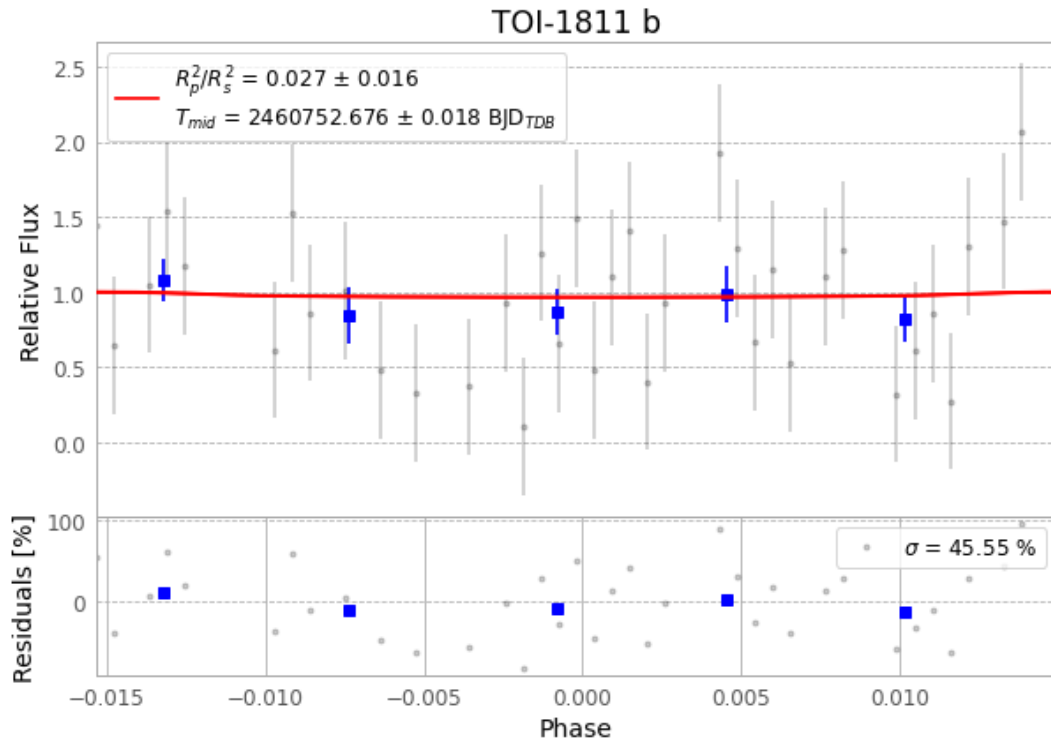


Figure 1 — FinalLightCurve_TOI-1811 b_2025-03-17.png (SHA-256 eefa79678e556066...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [369, 395], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5b0ffd4c7b350f34...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

55 FITS frames (36 MB), TOI1811250318024217.FITS → TOI1811250318052415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1811-250318.log (74231e96f20e...), FinalLightCurve_TOI-1811 b_2025-03-17.png (eefa79678e55...), FinalLightCurve_TOI-1811 b_2025-03-17.csv (0a4c71804e39...)

Compute resources

Wall time 164.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 23:37Z.